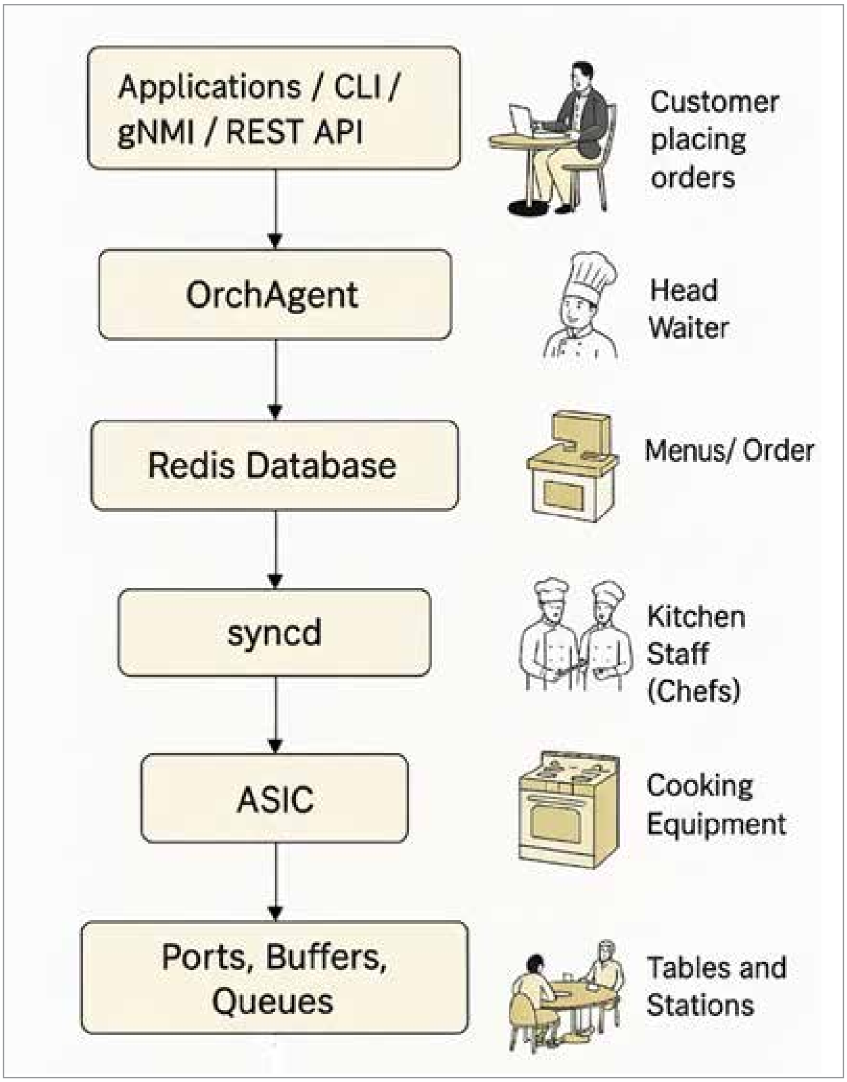

Figure 2: How configuration enters the SONiC stack (Image generated by ChatGPT)

queues, regardless of who built the chip. The hardware vendor supplies a SAI implementation; SONiC supplies the logic that calls it.

The result is portability. The same SONiC control plane can run on many different switching platforms, just as a chef can cook the same dish in multiple kitchens.

Serving packets at scale

Once the kitchen finishes preparing a dish, servers carry it to the table.

In networking terms, that is the data plane doing its job. Packets flow out of physical ports according to the state programmed earlier—routes selected, filters applied, quality-of-service enforced.

What makes SONiC distinctive is that the people taking orders, the kitchen manager, and the servers are all separate processes. Each runs in its own container and can be restarted or upgraded without shutting down the entire restaurant.

Why modular design matters

This architectural style delivers three major advantages.

First, it improves reliability. If one component misbehaves, it can often be restarted independently. Second, it encourages innovation. Developers can extend or replace modules without rewriting the whole system. Third, it keeps hardware choices flexible. Operators avoid being locked into a single vendor because the abstraction layer shields the upper software from chip-specific details.

Seeing SONiC in a new light

Analogies cannot replace technical documentation, but they help build intuition. Thinking of SONiC as a restaurant highlights its most important trait: cooperation between specialised workers through clear interfaces.

When orders flow smoothly from diners to chefs to servers, nobody notices the complexity underneath. That same invisibility is what operators want from their network operating systems—predictable behaviour, interchangeable components, and room to grow.

As SONiC adoption expands across the open source ecosystem, this modular approach is likely to remain its defining ingredient.

What to take away next

Understanding SONiC through an analogy is only the first step. If you are curious to dig deeper, explore how its databases coordinate state, how routing protocols plug in, and how vendors implement SAI for their chips.

Once you do, you may find that running a large data centre switch has more in common with running a fine restaurant than you ever expected. END

By: Rajshekhar Biradar

The author is a networking professional passionate about open networking systems, SONiC NOS, and AI fabric architectures. He enjoys breaking down complex systems into relatable stories that make technology easier to understand.